higher education & training

Department:
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REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N4

(8080164)

19 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

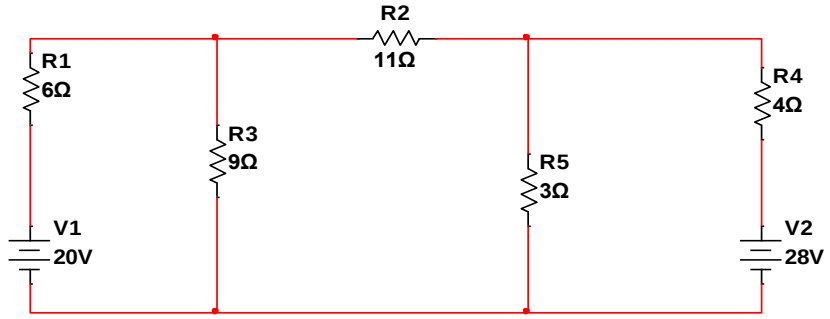
INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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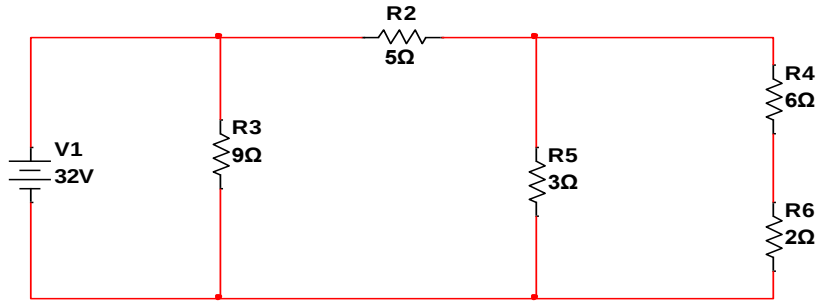
QUESTION 1

1.1 Study the circuit diagram below.



Calculate the value of the current flowing across the 3 Ω resistor using the superposition theorem. (6)

1.2 Study the circuit diagram below.



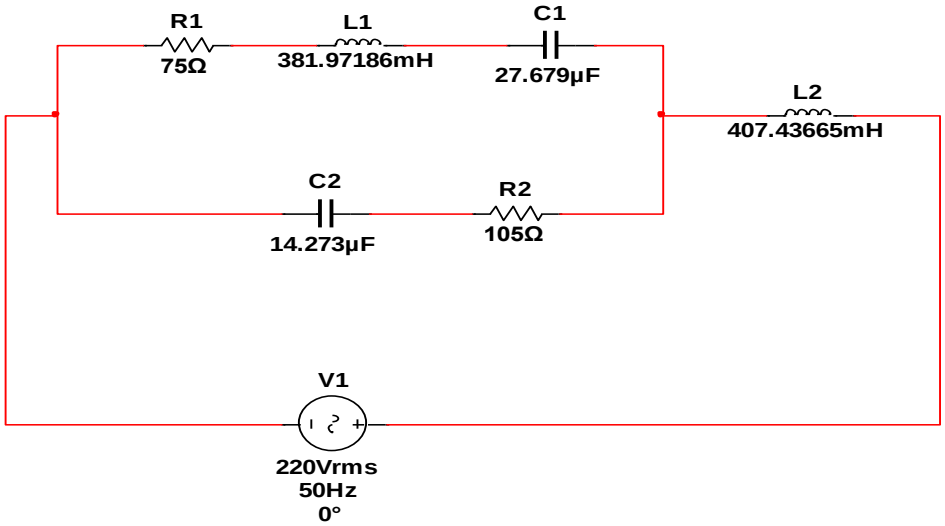
Calculate the current flowing across the 2 Ω resistor using Thevenin's theorem. (6)

[12]

QUESTION 2

2.1 Calculate the total bandwidth Δf and the edge frequencies f_H and f_L of an LC circuit resonating at 3 000 kHz with a Q of 200. (5)

2.2 Study the circuit diagram below.



Calculate the total current using the j notation. Assume the voltage supply to be $220 \angle 0^0 V$. (8)

[13]

QUESTION 3

3.1 Calculate the room temperature of a forward PN junction diode, which conducts 4 mA in the forward bias condition with a resistance of 7,233 Ω. Assume Boltzmann's constant 1,38 × 10^{-23} J/K and electron charge = 1,6 × 10^{-19}. (2)

3.2 Calculate the diode current of a silicon diode at 23 °C room temperature with I_s = 83 nF and an applied forward bias of 0,6 V. Assume Boltzmann's constant 1,38 × 10^{-23} J/K and electron charge = 1,6 × 10^{-19}. (2)

3.3 A Zener diode voltage regulator circuit has the following: V_{in} = 15 V, V_o = 8 V, P_2 = 5 W and I_L = 0,16 A.

Calculate the series resistor (R_S) and the load resistor. (6)

[10]

QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.1.1–4.1.4) in the ANSWER BOOK.

4.1.1 An autotransformer has only one winding, which serves as primary and secondary.

4.1.2 Rectification is the process of converting an alternating voltage to a unidirectional voltage.

4.1.3 A bridge circuit uses two diodes compared to the four used by the centre tap circuit.

4.1.4 In the LCπ filter, the output voltage is less than for the choke input filter.

(4 × 1) (4)

4.2 A step-down transformer 240/30 V at 50 Hz is connected to a rectifier, which in turn is connected to a capacitor filter. The capacitor has a value of 620 µF and the load resistor is 100 Ω. It is a half-wave rectifier.

Calculate the following:

4.2.1 Output voltage (V_{dc})

4.2.2 The RMS ripple voltage ($V_{r(rms)}$)

(2 × 4) (8)

[12]

QUESTION 5

5.1 Explain the following terms of a bipolar junction transistor (BJT):

5.1.1 Linear region

5.1.2 Cut-off region

5.1.3 Saturation

(3 × 2) (6)

5.2 Draw a complementary-symmetry push-pull class B. (4)

5.3 Explain the shunt feedback of the amplifiers. (2)
[12]

QUESTION 6

6.1 Draw a labelled circuit diagram of a non-inverting op-amp amplifier. (4)

6.2 An inverting op-amp has $R_f = 30\ k\Omega$, $V_{in} = 0{,}35\ V$, $V_o = 8\ V$.

Calculate R_{in} in kΩ. (3)

6.3 Draw a labelled circuit diagram of a summing amplifier. (4)
[11]

QUESTION 7

7.1 Explain each of the following applications of an SCR:

7.1.1 During the positive half-cycle (3)

7.1.2 During the negative half-cycle (2)

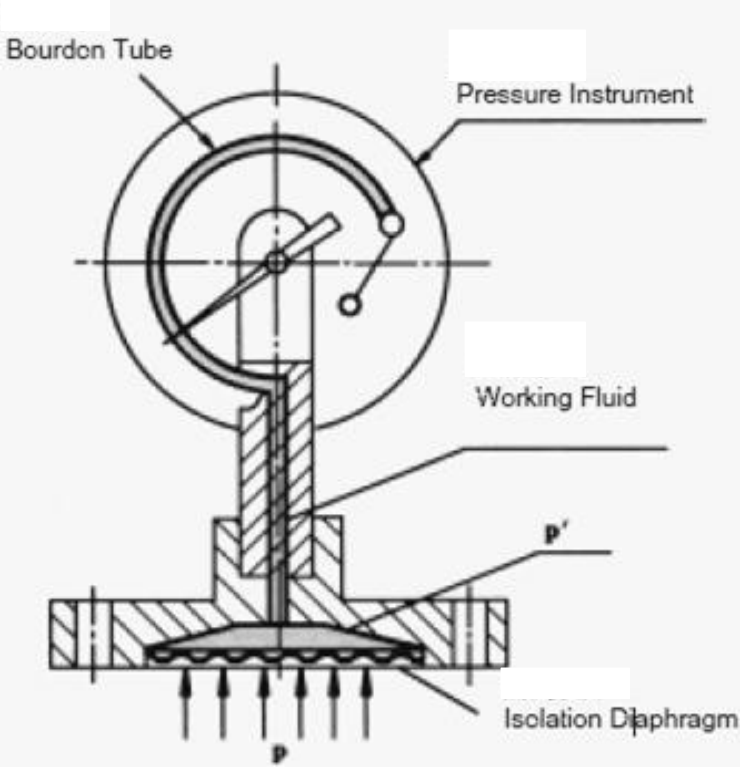
7.2 At what level of the current does the DIAC (diode for the alternating current) switch off? (2)

7.3 Explain the basic construction of a DIAC. (3)
[10]

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QUESTION 8

8.1 Study the circuit diagram below and explain its operating principle.



(5)

8.2 Draw a neat, fully labelled circuit diagram of a strain gauge. (3)

8.3 How many terminals does a phototransistor have? (2)

[10]

QUESTION 9

9.1 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (9.1.1–9.1.6) in the ANSWER BOOK.

COLUMN A		COLUMN B	
9.1.1	Samples the input waveform and produces a synchronising pulse	A	delay line
9.1.2	Delays arrival of input waveforms	B	intensity
9.1.3	Consists of high voltage to operate CRT	C	signal input
9.1.4	Determines brightness of spot on display	D	trigger circuit
9.1.5	Adjusts the amplitude of the waveforms	E	power supply
9.1.6	Connection of signal to be displayed	F	synchronisation
		G	volts/div

(6 × 1) (6)

9.2 An oscilloscope displays a time-base setting of 3 µsec/div, 0,6 V/div. 2,9 cm is measured from the 0 V reference line to the peak.

Calculate the following:

9.2.1 Peak voltage

9.2.2 V_{rms} and V_{ave}

(2 × 2) (4)

[10]

TOTAL: 100

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INDUSTRIAL ELECTRONICS N4

FORMULA SHEET

Any other applicable formula may also be used.

$$\frac{1}{R_T}=\left(\frac{1}{R_1}+\frac{1}{R_2}+...\frac{1}{R_n}\right) \qquad R_T=\frac{R_1R_2}{R_1+R_2} \qquad V_2=\frac{R_2}{R_1+R_2}\times\frac{V_T}{1}$$

$$Z=\sqrt{R^2+(X_L-X_c)^2} \quad Cos\,\theta°=\frac{R}{Z} \quad P=I^2R \quad P=\frac{V^2}{R} \quad P=VI\text{Cos}\,\theta$$

$$P=V\,.\,I \qquad F_r=\frac{1}{2\pi\sqrt{LC}} \qquad Q=\frac{X_L}{R} \qquad OF \;\; \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$I_t=\sqrt{I_R^2+(I_c-I_L)^2} \qquad Z=\frac{1}{\sqrt{\left(\frac{1}{R}\right)^2+\left(\frac{1}{X_c}-\frac{1}{X_L}\right)^2}} \qquad \frac{N_1}{N_2}=\frac{V_1}{V_2}=\frac{I_2}{I_1}$$

$$V_{rms}=0{,}707\,V_m \qquad i=I_s\left(e^{\frac{gv}{kT}}-1\right) \qquad R=\frac{kT}{qi} \qquad V.R=\frac{V_{NL}-V_{FL}}{V_{FL}}$$

$$V_{ave}=0{,}637\,V_m$$

$$f=\frac{1}{t} \qquad Rate\ of\ change=-\frac{V_{in}}{CR_{in}}$$

$$V_{dc}=0{,}318\,V_m \qquad\qquad V_{dc}=0{,}637\,V_m$$

$$\text{V}_{\text{r}_{\text{rms}}}=0{,}385\,\text{V}_\text{m}$$

$$PIV=V_m \quad \text{or} \quad 2\,V_m$$

$$V_{r_{rms}}=\frac{V_r\,(p-p)}{2\sqrt{3}} \qquad\qquad V_{dc}=V_m-\frac{V_r\,(p-p)}{2}$$

$$r=\frac{V_{r_{rms}}}{V_{dc}} \qquad\qquad V_{r_{rms}}=\frac{V_{dc}}{R_L\;2\sqrt{3}\;FC}$$

$$V_{dc}=V_m \quad \frac{I_{dc}}{2FC} \qquad\qquad r=\frac{I_{dc}}{V_{dc}\;2\sqrt{3}\;FC}$$

$$V_{r'_{rms}}=\frac{X_c}{\sqrt{R^2+X_c^2}}\times\frac{V_{r_{rms}}}{1}$$

$$V'_{dc}=\frac{R_L}{R_L+R_S}\times\frac{V_{dc}}{1} \qquad\qquad V_{r'_{rms}}=\frac{V_{r_{rms}}}{(2\pi f)^2\;LC}$$

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$$R_{in} = \frac{V_{be}}{I_b} \qquad R_{out} = \frac{V_{ce}}{I_c} \qquad R_c = \frac{V_{cc}}{I_c} \qquad V_{out} = R_1 C \frac{dv_i}{dt}$$

$$\textit{Static current gain} = \frac{I_{out}}{I_{in}}$$

$$\textit{Dynamic current gain} = \frac{\Delta I_{out}}{\Delta I_{in}}$$

$$V_{cc} = V_{RC} + V_{ce} \qquad V_{ce} = V_{cc} - V_{RC} \qquad R = \frac{p\ell}{a}$$

$$A_p = 10 \log \frac{P_{out}}{P_{in}} \qquad A_v = 20 \log \frac{V_{out}}{V_{in}} \qquad A_i = 20 \log \frac{I_{out}}{I_{in}}$$

$$\textit{Static voltage gain} = \frac{V_{out}}{V_{in}}$$

$$\textit{Dynamic voltage gain} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

$$hie = \frac{\Delta V_{in}}{\Delta I_{in}} = \frac{\Delta V_{be}}{\Delta I_b} \qquad V_{ce} = \textit{constant}$$

$$hre = \frac{\Delta V_{in}}{\Delta V_{out}} = \frac{\Delta V_{be}}{\Delta V_{ce}} \qquad I_b = \textit{constant}$$

$$hfe = \frac{\Delta I_{out}}{\Delta I_{in}} = \frac{\Delta I_c}{\Delta I_b} \qquad V_{ce} = \textit{constant}$$

$$hoe = \frac{\Delta I_{out}}{\Delta V_{out}} = \frac{\Delta I_c}{\Delta V_{ce}} \qquad I_b = \textit{constant}$$

$$V_{out} = \frac{R_f}{R_{in}} \times V_{in} \qquad V_{out} = -\left(\frac{R_f V_1}{R_1} + \frac{R_f V_2}{R_2} + \ldots \frac{V_n R_f}{R_n} \right)$$

$$V_{out} = \left(1 + \frac{R_f}{R_{in}} \right) \quad V_{in} \qquad V_{out} = -\frac{1}{CR_{in}} \int V_{in}\,(t)\,dt$$

Boltzmann's constant = 1,38 × 10^{-23} J/k

Electron charge = 1,6 × 10^{-19} C

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